

# HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

## PROFESSIONAL EXPERIENCE

**SOFTWARE ENGINEER – HCL Technologies | Chennai, India**

**09/2021 – 01/2025 (3.5 Years)**

- Architected and deployed **real-time networking infrastructure** for distributed access control systems, achieving a **99.9% uptime** under simulated unreliable network conditions.
- Developed **event-driven backend services** using **Python and AWS Lambda** to manage device <-> cloud communication, processing over **1 million device events daily**.
- Engineered **large-scale Linux device fleet management** solutions, automating deployment and monitoring for over **5,000 controllers** with **zero critical incidents**.
- Implemented robust **device identity, authentication, and security** protocols, reducing unauthorized access attempts by **80%** through certificate-based authentication.
- Designed and optimized **PostgreSQL schemas and queries** for platform systems, supporting **real-time data ingestion** and query response times under **100ms**.
- Enhanced **observability, debugging, and production tooling** using **Splunk and CloudWatch**, decreasing P1/P2 incident resolution time by **30%**.

## SKILLS

**Core Backend & Systems** – Distributed Systems, Real-time Networking, Event-Driven Architecture, Linux, PostgreSQL, Rust, Kubernetes, Cloud Infrastructure, Reliability Engineering, API Development.

**Programming Languages** – Python, Java, SQL, Bash Scripting.

**Cloud & DevOps** – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git.

**Data & Analytics** – Power BI, DAX, Tableau, R, pandas, NumPy, scikit-learn, Data Warehousing.

**Frameworks & Libraries** – Spring Boot, NestJS, Node.js, Flask, FastAPI.

**Methodologies & Tools** – Agile/Scrum, Jira, Confluence, Splunk, Dynatrace, Change Management.

## EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin**

**02/2026**

**BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India**

**01/2021**

## PROJECTS

**Real-time IoT Access Control Platform (HCL Technologies)**

- Developed a scalable backend platform for IoT access control systems, handling **real-time networking** and **device <-> cloud communication** for a fleet of 10,000+ Linux controllers.
- Implemented **distributed coordination and reliability** mechanisms using **PostgreSQL** and **Python**, ensuring consistent state management across devices and achieving **99.95% data integrity**.

- Engineered **device update orchestration and rollout systems** with automated rollback capabilities, successfully deploying updates to **95% of the fleet** within a 4-hour maintenance window.

#### **SWIFT Payments Operations Dashboard (HCL Technologies)**

- Supported a critical banking client's payments operations by enhancing a **SWIFT messaging dashboard**, improving transaction-flow visibility and operational alerting for **MT/MX messages**.
- Developed Python scripts for automated data validation and reporting on payment messages, reducing manual verification effort by **40%** and ensuring adherence to **ISO 20022 message structure**.
- Collaborated with business analysts to define requirements for **SEPA and cross-border payments** processing, contributing to a **15% reduction** in operational errors.

### **ACHIEVEMENTS & CERTIFICATIONS**

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)